

CodePilot – AI Coding Assistant: SQL Business Analysis Report

1. Introduction

CodePilot is an AI-powered coding assistant designed to support software development teams. Leadership wants to understand whether the current level of developer usage represents meaningful adoption, how usage differs across customers and teams, which features are used most, and whether product usage is related to business performance.

SQL is used to analyze developer activity, customer usage, feature adoption, usage trends, and revenue.

2. Business Objectives

The analysis aims to:

- Measure developer adoption.
- Compare usage across customers and teams.
- Identify the most frequently used features.
- Analyze usage trends over time.
- Compare product usage with customer revenue.
- Identify an additional business question for further investigation.

3. Dataset

The analysis uses a dummy dataset containing four related tables:

- **Customers** – customer details and subscription plans.
- **Developers** – developers associated with each customer and their teams.
- **Usage Events** – CodePilot usage activity and accepted suggestions.
- **Monthly Revenue** – monthly revenue generated from each customer.

Dataset Size

Table	Records
Customers	4
Developers	12
Usage Events	24
Monthly Revenue	12

The dataset covers **January to March 2026**.

The main relationships are:

Customer → Developers → Usage Events

and

Customer → Monthly Revenue

4. Assumptions

- A developer is considered Active if they have at least one usage event.
- A developer with no usage events is considered Inactive.
- Each usage event represents an interaction with CodePilot.
- Accepted suggestions represent AI suggestions accepted by developers.
- Revenue represents the monthly revenue generated from each customer.
- The dataset is dummy data created for SQL analysis.

5. Business Analysis

5.1 Developer Adoption

Question

How many developers are actively using CodePilot, and how many are inactive?

Solution

Each developer is checked against the usage events table. Developers with one or more usage events are classified as **Active**, while developers with no usage events are classified as **Inactive**.

SQL Concepts

CASE WHEN, COUNT(), LEFT JOIN, GROUP BY, Subquery

Findings

There are **10 active developers out of 12**, resulting in an overall adoption level of **83.33%**. Two developers are inactive.

5.2 Customer and Team Usage

Question

Which customers and teams use CodePilot the most?

Solution

Usage events and accepted suggestions are aggregated for each customer and team to compare their levels of CodePilot activity.

SQL Concepts

JOIN, LEFT JOIN, COUNT(), SUM(), COALESCE(), GROUP BY, ORDER BY

Findings

AlphaTech and CloudNova together account for **75% of all usage events**. Among teams, the **Platform** and **Data** teams have the highest usage.

5.3 Feature Usage

Question

Which CodePilot features are used most frequently?

Solution

Usage events are grouped according to their event type, such as code completion, chat, and code review. The number of events for each feature is then counted.

SQL Concepts

COUNT(), GROUP BY, ORDER BY

Findings

Code completion is the most frequently used feature, representing **66.67% of all usage events**.

5.4 Usage Over Time

Question

How does CodePilot usage change from month to month?

Solution

Usage events are grouped by month. The analysis measures the number of usage events, active developers, and accepted suggestions for each month.

SQL Concepts

DATE_FORMAT(), COUNT(), COUNT(DISTINCT), SUM(), GROUP BY, ORDER BY

Findings

January and February each recorded **10 usage events**, while March recorded only **4**. The number of active developers also decreased from **8 to 4** in March.

5.5 Usage vs Revenue

Question

Is higher CodePilot usage associated with higher customer revenue?

Solution

Total usage and total revenue are calculated separately for each customer and then compared.

SQL Concepts

JOIN, LEFT JOIN, Subqueries, COUNT(), SUM(), COALESCE(), GROUP BY

Findings

AlphaTech and CloudNova have relatively high CodePilot usage as well as high three-month revenue. However, the dataset does not establish that higher usage causes higher revenue. In March, usage decreased while total revenue continued to increase.

5.6 Additional Business Question

Question

Why did CodePilot usage decline sharply in March, particularly for some customers, while revenue continued to increase?

Solution

March usage can be compared across customers to identify which customers experienced the decline. Further investigation could examine developer engagement, feature adoption, training, product issues, or other customer-specific factors.

SQL Concepts

LEFT JOIN, COUNT(), GROUP BY, MONTH(), YEAR(), ORDER BY, Conditional filtering in JOIN

Findings

BetaWorks and DeltaSoft recorded **zero usage events in March**, while AlphaTech and CloudNova continued to use CodePilot. Therefore, the March decline is not uniform across all customers and requires further investigation.

6. Conclusion

The SQL analysis provides a business-oriented view of CodePilot adoption and usage. It identifies active developers, differences between customers and teams, popular features, monthly usage trends, and the relationship between usage and revenue.

The analysis also identifies a further business question: understanding the significant March usage decline, particularly for BetaWorks and DeltaSoft. Further customer and developer-level data would be required to investigate the reasons behind this change.

The SQL queries and their outputs are maintained separately in SQL Workbench for demonstration and review.